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Workforce training strategies and performance assessment in manufacturing environments: a preliminary investigation

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Abstract

Workforce skills greatly influence the production systems' performance and the companies' competitiveness. Training operators in the industry, particularly for complex non-routine activities like maintenance, is increasingly important. The scientific literature proposes several training, upskilling, and reskilling strategies using emerging technologies such as Virtual and Augmented Reality. However, a critical challenge remains in assessing the impact of training on performance and determining the best strategy to implement based on task and worker characteristics. Establishing the need for training and its effects on the system is still difficult. This study, starting from the lack of contextualization of the manufacturing training strategies filled by a framework proposal, aims to preliminary investigate the literature to detail training strategies and methods to evaluate the performance and skills achieved. The findings highlight a growing interest in smart and innovative training technologies and a lack of training economic evaluation according to the affected workers' performance.

Keywords: Workforce training; performance evaluation; human resource management; learning environment.

1. Background and motivation

Although the industrial environments are characterized by increasing digitalization and automation of processes, the workforce still remains the main asset element given its flexibility. Due to technological advances and changing demographics (such as the increasing working-age population), the global labor market is undergoing complex changes that require attention in industrial contexts. Understanding and adapting to the actual context is crucial for

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maintaining company's performance. The more the industrial environments move towards strong diversification of processes with mainly non-repetitive tasks to be performed (such as in the manufacturing of highly customized products or remanufacturing tasks that are not completely standardized), the more it will be necessary to train the workforce thoroughly to enhance skills and stimulate critical thinking, problem solving and adaptability [1, 2]. In this context, workforce training is crucial for the success of manufacturing enterprises, especially in the new era of Industry 5.0, as it ensures the preservation of productivity and competitive advantages. Industry 5.0 (I5.0) aims to transform the industrial sector into intelligent spaces through the integration of the Internet of Things (IoT), Artificial Intelligence (AI), and robotics. I5.0 emphasizes human-centricity, social sustainability, and sustainable development, redefining work dynamics and career paths [1–3]. A shift is required in the attitude toward employees, covering new requirements regarding workforce skills and new ways of attracting and retaining human capital equipped with adequate skills [2].

1.1. Reason for training, upskilling, and reskilling

The challenges in implementing workforce training, nowadays more than in the past, include the rapid pace of technological advancements and skill gaps to overcome. Significant investment in talent development emphasizing the workforce's upskilling and reskilling is needed [4]. In particular, upskilling refers to filling the skills gap created by technological advances enhancing the existing workforce skills through training programs, whereas reskilling involves acquiring completely new skills to adapt to the changing job profiles and the consequent skill requirements brought about by disruptive technologies in manufacturing environments [5]. As reported by Li in [6], the World Economic Forum estimates that by the next year (2025), half of all employees will need reskilling due to adopting new technology emphasizing the importance of lifelong learning. One of the reasons for this condition is that customers desire custom products that are adapted to their needs and requirements. This shifts the original mass production paradigm to the new mass customization one which requires greater flexibility and innovative industrial systems to respond rapidly to changes in production, new technologies, and product changes. According to [7], by 2030 approximately 2.1 million jobs in the United States are expected to remain unfilled due to a widening skills gap, worsened by ongoing technological advancements and shifting customer demands. As various surveys have shown, the shortage of skills is becoming an increasingly serious problem even in Europe. Focusing in particular on Italy, [8] confirms that the lowest-skilled occupations are those most exposed to the impact of technological innovation. The study highlights how the introduction of new technologies, such as Artificial Intelligence, does not lead to a direct replacement of the worker, but to a reorganization and redistribution of tasks among employed people. The result would be a reduction in labor input, but much less violent than that recorded with the analysis based on employed people. However, the risk of unemployment due to the adoption of new technologies would affect over 7 million employed people, or almost a third of the total.

1.2. Workforce training in manufacturing environments

The risk of technological unemployment is inversely proportional to the level of qualification of the occupation, which in turn is the result of the educational qualification held by the worker and the continuous training courses followed [8]. Considering the high costs associated with hiring new employees, manufacturers are increasingly focusing on training and upskilling their current workforce to adapt to the evolving manufacturing landscape. For these reasons, operators should be easily and effectively trained in this new era and use innovative tools to facilitate this action [9, 10]. On the other hand, however, conditions related to labor shortages and high turnover have made training of new employees a fundamental activity to ensure that workforce skills can be aligned to the needs of manufacturing environments. New employees, that potentially belong to the generation of inexperienced workers (Millennials and Gen Z) or come from the old generation that has decided to change due to the lack of good working conditions, present different sets of expectations and skills and for this reason, they need to be trained before performing well with a program that should adapt to their characteristics. Workforce training can be defined as a systematic development of the competencies needed by employees to perform their work [11]. To improve performance, such as time and quality, several methodologies can be identified based on the materials, techniques, and resources used to implement the training to transfer new knowledge, skills, and attitudes to the employees [12].

Training encourages internal growth improving productivity, reinforcing critical compliance and safety requirements, and potentially reducing employee turnover. In short, while training may be an expense for manufacturers, it is also an investment that they cannot afford to ignore to take competitive advantages [13]. Workplace training is an indispensable part of workforce skill development and can positively affect employees' safety knowledge, motivation, compliance, and participation, which are the main factors of safety performance [14, 15]. In the construction industry, lots of applications have been developed since the safety and efficiency of workers are two main issues that need to be addressed [16–18]. In other industrial contexts, such as in manufacturing, this seems not to be until now recognized as a relevant issue. However, this trend is shifting. Training programs in manufacturing are crucial for interacting with new technologies and ensuring productivity and competitive advantages, increasing flexibility, and mastering change in the production environment. Workplace training systems should consider didactic concepts, provide feedback and reflection mechanisms, and enable the integration of training and guidance for independent and effective learning [19].

Routine Tasks Training

For a long time, due to the carrying out of predominantly repetitive activities by operators, training was not considered a priority and fundamental in company choices and policies, nor strongly supported and encouraged by governments and academia except for mandatory training courses, mainly related to safety. For repetitive activities, i.e. actions that require the execution of the same actions repeated over time, often standardized and with well-defined guidelines and procedures, attention has focused on the study of learning curves. The learning curve is indeed a well-studied phenomenon with significant implications for performance improvement. A learning curve can be defined as a mathematical description of workers' performance in repetitive tasks. The more a task is repeated, the more the worker tends to perform the task in a shorter time due to its familiarity with the activities [20]. The learning effect is generally more focused on manual dexterity and consistent performance of specific tasks. The study by De Giorgio et al. has shown that different training methods, such as assembly animations or Augmented Reality, can influence the learning curves of operators [21]. The learning curve impacts skill retention in repetitive tasks through factors such as temporal distribution of practice, repetition schedule, and the nature of the task itself. High task repetitiveness was found to have a detrimental effect on well-being but increased work performance, highlighting the double-edged sword nature of repetitive tasks [22].

Non-routine tasks training

In the new manufacturing environment characterized by high variability, mass customization, and the use of automation and advanced technologies, the type of activities in which operators are involved are mainly non-repetitive. These activities can be defined as a set of operations that do not include processes or actions that are repeated over time, where the result is not pre-established and requires ingenuity, creativity, problem-solving skills, and the ability to adapt to different environments. Non-routine (or non-repetitive) tasks, due to their lower frequency, require both cognitive and manual skills. These activities, such as startup/shutdown activities or disposal of outdated products [23], by their nature, require operators to have a wide range of knowledge and skills to respond to unexpected and heterogeneous situations, and at the same time, a high level of quality is required. To guarantee these conditions, it is necessary to study and analyze how human beings learn new elements, and how their knowledge is maintained over time. Precisely because the operator's skills are requested infrequently, the disuse of a skill results in its degradation, and therefore in the lowering of the operator's level of preparation. In particular, the introduction of automation has placed greater cognitive loads (e.g., analysis, decisions, problem-solving, and judgment) on workers, while reducing physical loads. These tasks are the most difficult to manage for workers and the use of training for these specific tasks is still a challenge. While the repetitive task can benefit from the learning process, as described in the previous section, a significant aspect to consider in non-repetitive tasks is that they are characterized by a lower frequency taking a long time to re-perform the same activities for a single worker. This implies an accentuated forgetting process, i.e., the decline of memory retention in time due to the information lost over a period when there is no attempt to retain it, and the task is not carried out. Due to the characteristics of these tasks, the possibility of making errors, injuring, or lowering performance increases significantly since employees have performed the task in the past, but not recently, they might not remember the proper procedure to follow [23]. More and more workforce training is required for non-repetitive tasks and this approach completely differs from that for repetitive tasks due to the need for increased flexibility and adaptability in the production environment. Nowadays the key problem in providing training for non-repetitive tasks in manufacturing is efficiently training personnel for an increasing and dynamic range of tasks while adapting training measures to individual skill levels and task requirements. Designing successful hands-on training programs for non-repetitive tasks involves arranging for the transfer of knowledge in production environments and systematically guiding the conception of hands-on training. Furthermore, the use of digital learning games and digital learning components can supplement traditional learning factories, always enabling individual learning routes for all trainees and accessible learning and places [24]. The need for a high level of employee qualification across all areas of the company is essential to cope with the challenges posed by non-repetitive manufacturing processes is critical.

An emerging aspect, in the context of workforce training, is the use of digital technologies, such as Virtual and Augmented Reality (VR and AR) environments, which offer a possible solution to train workers in an immersive, interactive, and safe environment, enabling the development of a curriculum for workers in the industrial manufacturing sector guarantying efficient and effective learning process [25]. Additionally, mixed reality applications and adaptive task tutoring systems have gained attention for training unskilled workers in smart factories, providing effective and independent training for new employees [26]. The use of prescriptive training strategies, game-based training systems, and evolutionary-based fuzzy cognitive maps can schedule training sessions, recreate virtual industrial contexts, and extract workers' implicit procedural knowledge to facilitate knowledge transfer and sustain workforce competence development [23].

1.3. Research study aim

In the current industrial dynamic scenario, although the need to train the workforce is clear and technology can improve the methodologies to reach the workers, the attention of researchers has so far not focused on defining an overall framework that can consider the entirely logical and procedural flow that is needed for training courses and its impact on the overall performance of the system. Most research studies reviewed the literature focusing on specific technologies used in training [27, 28] or proposing single case study applications [10, 29–31]. Furthermore, the literature review on the training strategies proposed by Braun et al. (2024) focused mainly on the involved stakeholders and their collaboration from a managerial more than a practical point of view [32] whereas the research study by Cazeri et al. (2022) [33] explored training for the Industry 4.0 also in this case investigating the different stakeholder involved. However, to the best of the authors' knowledge, there are no studies focused on the strategies and programs developed for workforce training in manufacturing environments and the effect in terms of assessment of competencies and performance on the overall system. Training has been recognized as fundamental but until now no systematic approach that includes this task and its main peculiarities has been proposed. This activity is still not centralized and not fully connected to the entire manufacturing system in which and for which it is needed. For these reasons, the main objective of this study is to focus on and detail the main strategies used in the literature for training the workforce in manufacturing environments, including different types of technologies, and to establish the methods currently used to assess the skills and competencies achieved integrating them in framework proposal. The Research Questions (RQs) that this study wants to address are:

- RQ1: What are the main characteristics of programs developed for training in manufacturing environments?
- RQ2: How to measure the effect of training in terms of performance and skills?

This article is organized as follows. In Section 2 the methodology that this research study followed is presented, whereas the results of our analysis are detailed in Section 3. In Section 4 a brief discussion follows, and the main conclusions of this work are drawn.

2. Methodology

To adequately address the above-mentioned RQs, this study is built on a deductive research approach in which, starting from the knowledge represented by extant literature and mediated by the authors' direct experience in an industrial context concerning the effects of the lack of experience and training of the operators, a preliminary training framework is proposed. The scientific literature on operator performance in industrial contexts has been integrated with secondary data (e.g., white papers, consulting company reports, focus groups with industrial experts) analysis to understand the most critical aspects of the low level of expertise of operators. Based on this evidence, the training protocols currently employed in the industrial field have been analysed to identify the most used structure and the main problems. Since several challenges convey the workplace training systems, including the need for didactic concepts to enable and verify learning and retention of acquired skills, the provision of feedback mechanisms for long-term user motivation, and the integration of training and guidance for independent work by new employees; this study focused on the two aspects of strategies and performance assessment. Through an overview of the scientific literature, carried out on the Scopus database and focusing on the last 10 years, the main objective that this study wants to achieve is to detail the main strategies proposed in the literature for training the workforce in manufacturing environments, including and considering different types of technologies, and to establish the methods currently used to assess the skills and competences achieved through training by operators.

3. Framework proposal

In tasks that involve human employees, the lack of training, and, in general, of experience in case of non-repetitive tasks, affects industrial performance in terms of quality, productivity, and safety [11, 13, 14, 34]. Inadequate training can lead to a multitude of issues that have far-reaching consequences for both the employees and the overall processes. Research studies revealed that the Key Performance Indicators (KPIs) affected by training processes are completion time and task error detected [10, 29, 35] and other subjective metrics like motivation and satisfaction [29, 35]. Firstly, quality is often compromised when employees are not properly trained. This can cause product defects, which range from minor issues to major functional flaws [34]. In this optics, inconsistent quality control can damage the company's reputation, leading to customer dissatisfaction. Secondly, productivity suffers when the workforce lacks the necessary skills and knowledge. Untrained or in general unexperienced employees are more likely to make errors and miss crucial steps in the production process, leading to delays and inefficiencies. Setting up equipment incorrectly or operating machinery improperly can further slow production, reducing overall output and increasing operational costs. The preliminary framework on which this work is based is reported in Fig. 1.

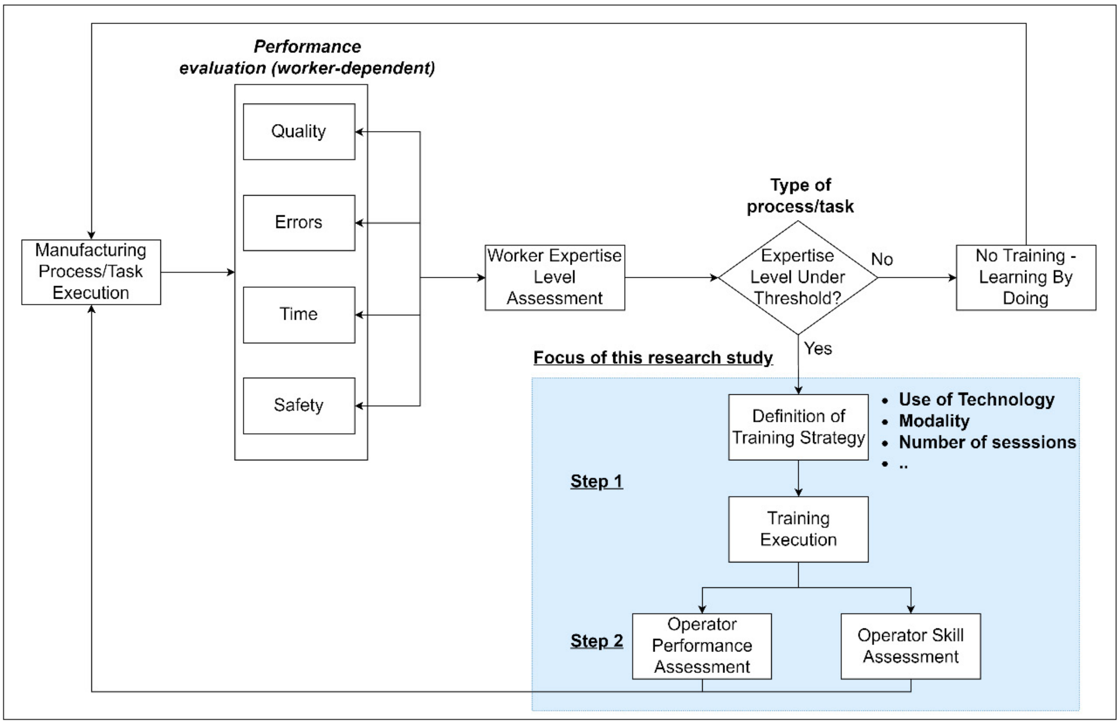

Fig. 1 Preliminary Training Framework in Manufacturing Environments.

It defines when a training session may be needed, according to the decreased performance of operators, and what are the main elements to follow in the execution of the training program. Starting from the operator-based process, key performances that could be affected by a lack of training or in general a lack of skill should be investigated and quantified. To evaluate the performance of the operators, KPIs have been selected: measurable, they must be able to be quantified in a clear and objective way; actionable, they must provide insights that can be used to undertake corrective actions; relevant, they must reflect the company's objectives. For example, for each type of task, performance such as the quality of the product, time spent on the task, the number of errors and unsafe incidents or conditions should be assessed. By combining single performance over time, it should be possible to establish an overall level of expertise for the worker. Given the specific context of the task, if a critical threshold for the overall expertise level is identified (as the overall performance decreases), this allows to determine the need for operator training. If the threshold level is not reached, the operator performs the activity without a prior training process. In this case, the direct experience allows skills to be maintained by triggering indirect learning by performing the process, or in other words, the learning experience results from the operator's actions. In general, knowledge at the individual level accumulates by doing tasks because each time a task is realized, the experience allows one to increase the personal levels of skills and competence on the specific and correlated tasks. At this point, the best strategy should be chosen for the individual operator, both in terms of strategy (e.g., training time, number of training sessions, etc.) and modality (e.g., online lessons, in-person lessons, immersive environments, use of augmented reality, etc.). The strategies and modalities are closely interrelated. Once the training has taken place, the overall level of skills acquired by the operator should be quantified and made available in some way also for other kinds of tasks. Improvement in overall performance will be ensured for individual tasks. Different tasks may require similar skills, and the operator's overall level of training can be utilized in this context. The main findings related to the training strategies (Step 1 in Fig. 1) and assessment methods (Step 2 in Fig. 1) will be presented in the next sections and represent the focus of this research work.

3.1. Programs for training in manufacturing

A manufacturing training program is a structured educational initiative aimed at providing employees with the skills and knowledge essential for productivity and safety in their manufacturing roles. This program covers various topics, such as safety protocols, equipment operation, quality control, and industry-specific processes. Given a specific process (such as assembly, maintenance, etc.), the training programs can differ for:

- ***Modality*** (e.g., type of technology involved, paper-based or not, etc.)
- ***Strategy*** (e.g., training time, number of training sessions, etc.)

For sure, the modality affects the choice in terms of strategy given the effect of technologies on workers. Innovative technologies such as Virtual Reality (VR) [9, 10, 16, 29, 36], Augmented Reality (AR) [31, 37, 38], mixed reality (XR) [19, 29, 39], and gaming [40] environments are being explored to enhance training, offering the potential to overcome the limitations of traditional approaches.

The drawbacks of traditional training methods include the limited, non-holistic perspective of each trainee, non-recurring knowledge transfer, and the need for trainees to be in the same place at the same time. Technologies help overcome these limitations and can be used in different cases. For example, as suggested by [39], VR environments can be more useful for the earlier stage of the training program, when the workers/employees do not have enough knowledge and experience of the real environment, whereas AR can support more advanced stages of the training program when the trainees only need support. This point opens a significant issue related to the influence of operator characteristics (such as experience but not only) in the choice of the modality to follow. Training VR applications can increase the efficiency of procedural knowledge transfer. It is much easier to repeat tasks, and no dangerous situations are provided. Moreover, VR environments present advantages such as the scalability and adaptability of the content, which can be adapted to specific users and changes in conditions in the simulation process [10].

Augmented Reality, which adds augmented information in the real world, has been shown to improve learnability by workers [31], even if less investigated for training applications in comparison to VR. Augmented reality seems to be more practically used for assistance applications. The classical approaches such as paper-based or in-person, even if traditionally useful, are not investigated as the main topic by researchers but are only used for comparison. For example, in-person training is still today effective but requires the time and effort of experts without the possibility of scaling the application to other workers [26].

Further approaches for training include traditional Classroom Training, with theoretical teachings from experts and the use of detailed guides and manuals to support learning, online courses with e-learning platforms with interactive modules and videos or on-the-job Training (OJT), a method of training where employees learn how to perform their job duties while working. Instead of learning in a classroom setting or through simulations, trainees acquire skills and knowledge in the actual work environment.

Regarding the strategies, and so the overall duration of the training task, distribution of training activities over time, etc., no standard approach can be defined. Each research study, according to the methodology chosen, provides its main definition. Moreover, performance and skills change according to the training process, and this should also be considered in a double way. It is interesting to highlight that training activities, which are vital and useful, on the one hand, employ worker availability, and this must be considered in the overall planning of tasks, while on the other, they affect the future performance of the systems.

3.2. Assessment method for workforce training

According to the type of training assessment method to follow, specific data collection tools are required. The most common methods for evaluating the manufacturing training of the workforce include questionnaires, interviews, focus group discussions, tests, observations, and performance records [41]. These methods impact the skill development of the manufacturing workforce by providing tailored approaches for assessing competencies and training efficacy.

In general, the assessment can be defined according to the performance and the competencies, even if this case is more complex to manage and collect data.

The assessment of competencies in manufacturing environments is crucial for success in adapting to changing market demands and global competition. An approach using a structural equation analysis is presented by Rupert and Metternich [42] to determine the acquisition of competencies in learning factories. This approach aims to develop a statistically sound measurement procedure by using diverse parameters such as written tests, observation, and self-assessment. Regarding instead the performance, a study on the influence of the Employee Training and Assessment Register (TAR) on productivity in an automotive parts manufacturing organization in South Africa found that spoilage rate has a relationship with company productivity, making it a potential KPI for assessing workforce training [43].

In several studies, such as in the studies [10, 29, 35], the comparison among different training methodologies is mainly based on the measurement of performance such as the time involved in the task and the number and type of errors. Workforce training in manufacturing significantly impacts operator performance in terms of productivity and efficiency [34]. The quality of production, performance of deliveries, and overall success of the business depends largely on the skills of the operators in the manufacturing industry. Computer-based training of operational sequences and related quality information has been shown to support the assembly performance of operators, leading to clear positive differences in learning progress and improvements in quality output [34]. The effectiveness of workforce training can be measured using metrics such as learning progress, output, and productivity gains.

4. Discussions and Conclusions

In highly variable and dynamic manufacturing environments, characterized by increasing levels of technology and automation, the role of the worker is still fundamental, and the ability and skills of workers need to be enhanced to ensure the efficiency of processes, safety, and quality. The lack of proper training can lead, in fact, to significant adverse effects, which can impact various aspects of the business performance. Only to provide some examples, without training, workers may take longer to complete their tasks and make more mistakes. This inefficiency can lead to slower production times, higher operational costs, and decreased overall productivity. Proper training ensures that employees understand the standards and procedures necessary to produce high-quality products and avoid injuries or fatalities due to the lack of the right execution of procedures [12, 34]. The matter of training programs is strictly related also to the complexity of the task to perform. For example, VR training, given its contribution to better perception and faster identification of task details, has been considered better than other strategies for learning complex tasks more than simple ones [10]. Furthermore, case studies investigated and proposed in the literature are generally based on experiments in the laboratory and this could significantly affect the results obtained. The real workforce characteristics cannot be fully covered by university students although they represent a solution for testing and preliminary evaluating training approaches. Although the analysis in this paper focused on training techniques in the manufacturing sector, i.e. for personnel already in the working world, the analysis carried out revealed that employers and educational institutions must make joint efforts to ensure the development of quality professional education and reduce the gap between employee qualifications and graduates'/scholars' competences [44–46]. In conclusion, this research work wants to provide a preliminary investigation of the scientific literature on the different training strategies performed in the actual industrial environments focusing on the evaluation of the performance/skills achieved acquired in industrial contexts and assessed.

4.1. Future research agenda

Training operators effectively requires careful consideration of when and how to conduct training sessions tailored to individual characteristics. Instead of a one-size-fits-all approach, it is crucial to assess factors such as skill level, experience, and learning preferences. This personalized approach ensures training is meaningful and impactful. Strategies could involve initial assessments, continuous monitoring, and feedback loops to adapt training interventions over time. Regarding technology instead, Virtual Reality environments offer immersive and interactive settings that enhance both soft and hard skill training for operators. The advantage lies in providing a contextually rich training experience that bridges the gap between theory and practice providing safe environments in which operators can learn.

While assembly and manufacturing processes are more technical and are easier to study and investigate [25, 29, 38], fields such as maintenance and disassembly present unique challenges. These tasks often involve higher variability and complexity and require problem-solving abilities. Training in these areas should emphasize adaptive thinking, troubleshooting skills, and the ability to respond to unexpected situations. Developing training modules that replicate real-world scenarios can better prepare operators for these dynamic environments.

There seems to be a lack of studies evaluating the effects of training on operators from an economic point of view. The cost-effectiveness of training strategies is not evaluated from a managerial point of view, and this is strictly connected to the choice of the best strategy and modality of training approach and the evaluation of the effect of training on the future performance of the system. The development from a technological point of view of the training environments and contents should affect significantly the economic impact of training. However, it is fundamental to focus on the results of training tasks and their effectiveness and effect on the performance to propose economic model.

In addition, there is a notable lack of standardized methods for assessing and evaluating operator skills across manufacturing systems. This inconsistency makes it difficult to measure proficiency objectively and use assessment data effectively for manufacturing system improvements. Implementing robust evaluation frameworks, such as competency-based assessments or performance metrics can provide clearer insights into operator capabilities. These insights can guide targeted training programs and support continuous improvement initiatives within the manufacturing environment. To enhance overall manufacturing system performance, it is essential to leverage assessment data strategically. By aligning training outcomes with operational goals, organizations can identify skill gaps, optimize training investments, and foster a culture of continuous learning. Regular feedback loops between assessment results and training interventions enable iterative improvements that align with evolving industry standards and technological advancements.

4.2. Limitations of this study

Given the breadth of the topic and the initial exploration, an overview of the existing literature has been carried out. It is important to acknowledge the challenge of conducting a systematic review to highlight key theories, trends, and significant findings, while also noting the gaps or emerging areas for further research more systematically. It should be interesting to review, given a single field of application, all the training methods and strategies developed and clearly define the effect in terms of performance and skill. Moreover, it should be interesting to focus on case study applications in real industrial environments to highlight the main challenges and problems in practical implementations of training programs.

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References

[1] Romero, David, and Johan Stahre. (2021) "Towards the Resilient Operator 5.0: The Future of Work in Smart Resilient Manufacturing Systems." *Procedia CIRP* **104** (**March**): 1089–94. Available from: https://doi.org/10.1016/j.procir.2021.11.183

[2] Lagorio, Alexandra, Chiara Cimini, and David Romero. (2023) "A Reflective Framework for Understanding Workforce Evolutionary Pathways in Industry 5.0." *IFIP Advances in Information and Communication Technology* **689 AICT** (**September**): 501–12.

[3] Maddikunta, Praveen Kumar Reddy, Quoc Viet Pham, Prabadevi B, N. Deepa, Kapal Dev, Thippa Reddy Gadekallu, et al. (2022) "Industry 5.0: A survey on enabling technologies and potential applications." *Journal of Industrial Information Integration* **26** (**June 2021**).

[4] Gamberini, Luciano, and Patrik Pluchino. (2024) "Industry 5.0: A comprehensive insight into the future of work, social sustainability, sustainable development, and career." *Australian Journal of Career Development* **33** (**1**): 5–14.

[5] Pontes, Joseane, Carla A.S. Geraldes, Florbela P. Fernandes, Lucas Sakurada, Ann Lilith Rasmussen, Lasse Christiansen, et al. (2021) "Relationship between Trends, Job Profiles, Skills and Training Programs in the Factory of the Future." *Proceedings of the IEEE International Conference on Industrial Technology* **2021-March** : 1240–5.

[6] Li, Ling. (2022) "Reskilling and Upskilling the Future-ready Workforce for Industry 4.0 and Beyond." *Information Systems Frontiers* (**0123456789**).

[7] Wellener, Paul, Heather Ashton, Victor Reyes, and Chad Moutray. (2021) "Creating pathways for tomorrow's workforce today: Beyond reskilling in manufacturing." [Internet]Deloitte Insights. Available from: https://www2.deloitte.com/us/en/insights/industry/manufacturing/manufacturing-industry-diversity.html/#the-growing

[8] Tecnica, Assistenza, Fondo Formazione Italia, Strumenti D.I. Finanziamento, and Fondo Trasparente. (2022) "I riflessi dell ' innovazione tecnologica sul mercato del lavoro italiano."

[9] Karagiannis, Panagiotis, Theodoros Togias, George Michalos, and Sotiris Makris. (2020) "Operators Training Using Simulation and VR Technology." *Procedia CIRP* **96** (**March**): 290–4.

[10] Zawadzki, Przemyslaw, Krzysztof Zywicki, Pawel Bun, and Filip Gorski. (2020) "Employee Training in an Intelligent Factory Using Virtual Reality." *IEEE Access* **8** : 135110–7.

[11] Castillo-Apraiz, Julen, and Jesús Matey De Antonio. (2020) "The mediating role of personnel training between innovation and performance: Evidence from the German pharmaceutical industry." *Cuadernos de Gestion* **20** (**3**): 41–52.

[12] Ibrahim, Rosli, Ali Boerhannoeddin, and Kazeem Kayode Bakare. (2017) "The effect of soft skills and training methodology on employee performance." *European Journal of Training and Development* **41** (**4**): 388–406.

[13] Arulsamy, A.S., Indira Singh, M. Senthil Kumar, Jetal J. Panchal, and K.K. Bajaj. (2023) "Employee Training and Development Enhancing Employee Performance – A Study." *Samdarshi* **16** (**3**): 1–11.

[14] Tappura, Sari, Aki Jääskeläinen, and Julius Pirhonen. (2021) "Performance Implications of Safety Training." *Lecture Notes in Networks and Systems* **262 LNNS** : 295–301.

[15] Hutchinson, Derek, Gil Luria, Shani Pindek, and Paul Spector. (2022) "The effects of industry risk level on safety training outcomes: A meta-analysis of intervention studies." *Safety Science* **152** (**April 2021**).

[16] Roofigari-Esfahan, Nazila, Curt Porterfield, Todd Ogle, Tanner Upthegrove, Myounghoon Jeon, and Sang Won Lee. (2022) "Group-based VR Training to Improve Hazard Recognition, Evaluation, and Control for Highway Construction Workers." *Proceedings - 2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, VRW 2022* : 513–6.

[17] Jadallah, Hazem, Carol J. Friedland, Isabelina Nahmens, Charles Pecquet, Charles Berryman, and Yimin Zhu. (2021) "Construction Industry Training Assessment Framework." *Frontiers in Built Environment* **7** (**August**): 1–12.

[18] Voordijk, Hans, and Farid Vahdatikhaki. (2020) "Virtual Reality learning environments and technological mediation in construction practice." *European Journal of Engineering Education* : 1–15. Available from: https://doi.org/10.1080/03043797.2020.1795085

[19] Pimminger, Sebastian, Werner Kurschl, and Johannes Schonbock. (2022) "Mixed Reality Workplace Training Systems for Smart Factories: Challenges and Future Directions." *Proceedings of the 2022 IEEE International Conference on Human-Machine Systems, ICHMS 2022* (**iii**).

[20] Anzanello, Michel Jose, and Flavio Sanson Fogliatto. (2011) "Learning curve models and applications: Literature review and research directions." *International Journal of Industrial Ergonomics* **41** (**5**): 573–83.

[21] de Giorgio, Andrea, Stefania Cacace, Antonio Maffei, Fabio Marco Monetti, Malvina Roci, Mauro Onori, et al. (2022) "Assessing the influence of expert video aid on assembly learning curves." *Journal of Manufacturing Systems* **62** (**June 2021**): 263–9.

[22] Häusser, Jan Alexander, Stefan Schulz-Hardt, Thomas Schultze, Anne Tomaschek, and Andreas Mojzisch. (2014) "Experimental evidence for the effects of task repetitiveness on mental strain and objective work performance." *Journal of Organizational Behavior* **35** (**5**): 705–21.

[23] Longo, Francesco, Antonio Padovano, Fabio De Felice, Antonella Petrillo, and Mohaiad Elbasheer. (2023) "From 'prepare for the unknown' to 'train for what's coming': A digital twin-driven and cognitive training approach for the workforce of the future in smart factories." *Journal of Industrial Information Integration* **32** (**February**).

[24] Görke, Matthias, Vivian Bellmann, Jan Busch, and Peter Nyhuis. (2017) "Employee Qualification by Digital Learning Games." *Procedia Manufacturing* **9** : 229–37. Available from: http://dx.doi.org/10.1016/j.promfg.2017.04.040

[25] Ho, Nicholas, Pooi Mun Wong, Matthew Chua, and Chee Kong Chui. (2018) "Virtual reality training for assembly of hybrid medical devices." *Multimedia Tools and Applications* **77** (**23**): 30651–82.

[26] Huang, Gaoping, Xun Qian, Tianyi Wang, Fagun Patel, Maitreya Sreeram, Yuanzhi Cao, et al. (2021) "Adaptutar: An adaptive tutoring system for machine tasks in augmented reality." *Conference on Human Factors in Computing Systems - Proceedings*.

[27] Checa, David, and Andres Bustillo. (2020) "A review of immersive virtual reality serious games to enhance learning and training." *Multimedia Tools and Applications* **79** (**9**–**10**): 5501–27.

[28] Knoke, Benjamin, and Klaus Dieter Thoben. (2021) "Training simulators for manufacturing processes: Literature review and systematisation of applicability factors." *Computer Applications in Engineering Education* **29** (**5**): 1191–207.

[29] Jalilvand, Iman, Jiyoung Jang, Bhushan Gopaluni, and Abbas S. Milani. (2024) "VR/MR systems integrated with heat transfer simulation for training of thermoforming: A multicriteria decision-making user study." *Journal of Manufacturing Systems* **72** (**December 2023**): 338–59.

[30] Maettig, Benedikt, and Hermann Foot. (2020) "Approach to improving training of human workers in industrial applications through the use of Intelligence Augmentation and Human-in-the-Loop." *15th International Conference on Computer Science and Education, ICCSE 2020* (**Iccse**): 283–8.

[31] Ariansyah, Dedy, Bens Pardamean, Eddine Barbaro, and John Ahmet. (2024) "Augmented reality training for improved learnability." *CIRP Journal of Manufacturing Science and Technology* **48** (**March 2023**): 19–27.

[32] Braun, Greta, Pauliina Rikala, Miitta Järvinen, Raija Hämäläinen, and Johan Stahre. (2024) "Bridging Skill Gaps - A Systematic Literature Review of Strategies for Industry." *Advances in Transdisciplinary Engineering* **52** (**April**): 687–96.

[33] Cazeri, Gustavo Tietz, Luis Antonio De Santa-Eulalia, Milena Pavan Serafim, and Rosley Anholon. (2022) "Training for Industry 4.0: A systematic literature review and directions for future research." *Brazilian Journal of Operations and Production Management* **19** (**3**): 1–19.

[34] Malmsköld, Lennart, Roland Örtengren, and Lars Svensson. (2015) "Improved quality output through computer-based training: An automotive assembly field study." *Human Factors and Ergonomics In Manufacturing* **25** (**3**): 304–18.

[35] Passalacqua, Mario, Robert Pellerin, Esma Yahia, Florian Magnani, Frédéric Rosin, Laurent Joblot, et al. (2024) "Practice With Less AI Makes Perfect: Partially Automated AI During Training Leads to Better Worker Motivation, Engagement, and Skill Acquisition." *International Journal of Human-Computer Interaction* **0** (**0**): 1–21. Available from: https://doi.org/10.1080/10447318.2024.2319914

[36] Caterino, Mario, Marta Rinaldi, Valentina Di Pasquale, Alessandro Greco, Salvatore Miranda, and Roberto Macchiaroli. (2023) "A Human Error Analysis in Human–Robot Interaction Contexts: Evidence from an Empirical Study." *Machines* **11** (**7**).

[37] Vidal-Balea, Aida, Oscar Blanco-Novoa, Paula Fraga-Lamas, Miguel Vilar-Montesinos, and Tiago M. Fernández-Caramés. (2020) "Creating collaborative augmented reality experiences for industry 4.0 training and assistance applications: Performance evaluation in the shipyard of the future." *Applied Sciences (Switzerland)* **10** (**24**): 1–23.

[38] Büttner, Sebastian, Michael Prilla, and Carsten Röcker. (2020) "Augmented Reality Training for Industrial Assembly Work - Are Projection-based AR Assistive Systems an Appropriate Tool for Assembly Training?" *Conference on Human Factors in Computing Systems - Proceedings* : 1–12.

[39] Santos, Jose Eduardo, Miguel Nunes, Marcelo Pires, Jose Rocha, Nuno Sousa, Telmo Adao, et al. (2022) "Generic XR game-based approach for industrial training." *ICGI 2022 - International Conference on Graphics and Interaction, Proceedings*.

[40] Ulmer, Jessica, Sebastian Braun, Chi Tsun Cheng, Steve Dowey, and Jorg Wollert. (2020) "Gamified Virtual Reality Training Environment for the Manufacturing Industry." *Proceedings of the 2020 19th International Conference on Mechatronics - Mechatronika, ME 2020*.

[41] Kucherov, Dmitry, and Daria Manokhina. (2017) "Evaluation of training programs in Russian manufacturing companies." *European Journal of Training and Development* **41** (**2**): 119–43.

[42] Glass, Rupert, and Joachim Metternich. (2020) "Method to measure competencies - A concept for development, design and validation." *Procedia Manufacturing* **45** : 37–42.

[43] Zondo, Robert W.D. (2019) "The impact of training and assessment register (TAR) on productivity in the automotive parts manufacturing organisation in South Africa." *SA Journal of Human Resource Management* **17** : 1–8.

[44] Putilov, Alexandr V., Mikhail N. Strikhanov, and Georgy V. Tikhomirov. (2019) "Personnel training for the developing nuclear power industry." *Nuclear Energy and Technology* **5** (**3**): 201–6.

[45] Iskakov, I.Z., B.B. Kovalenko, M.S. Turovskaia, and G. V. Getmanova. (2020) "Online blended learning in the digital environment." *Journal of Physics: Conference Series* **1691** (**1**).

[46] Ismoilova, Gulnora, Dildora Khakimdjanova, Muazzamkhon Shaislamova, and Feruza Nabieva. (2020) "Prospects for the training of highly qualified personnel in the modernization economy." *International Journal of Scientific and Technology Research* **9** (**3**): 3629–34.